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Characterisation of the flow behaviour of pharmaceutical powders using a model die–shoe filling system

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Abstract

In the pharmaceutical industry it is important to understand the factors that influence the flowability of a powder and to identify suitable experimental procedures which can be used to evaluate powders and guide the selection of process parameters. This paper determines the flowability of seven pharmaceutical powders using a model die–shoe filling rig. Dimensional analysis is employed to provide a framework for the interpretation of experimental results and to guide the development of procedures to extrapolate the results to other die and shoe geometries. It is demonstrated that the variation of mass delivered into a standard die as a function of shoe velocity provides a measure of flowability. Qualitative and quantitative studies were undertaken using a high-speed video system and by determining the critical shoe velocity, i.e. the shoe velocity above which incomplete filling occurs.
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Keywords: Measurement of powder flow; Die–shoe filling rig; Critical velocity; High-speed video; Rotary press

1. Introduction

The flow properties of a powder are important in pharmaceutical tablet manufacturing as most processes, such as mixing, granulation, drying, milling and compaction involve powder flow. The transfer of powders between these unit operations involves flow in hoppers, pipes, chutes etc. The flow of powders can be divided into 1) slow (i.e. hopper flow) where energy is dissipated through inter-particle friction [1,2] and 2) rapid flow (i.e. avalanches) which are dominated by collisions between particles [3]. Parameters such as dimensionless shear rate [4] were proposed to describe various flow regimes. During a manufacturing process, these different types of flow can occur concurrently, with the details of the flow behaviour influenced by the loading experienced by the particles (gravity, vibration, body forces, interaction with air flow and the container boundaries etc.), the fundamental powder characteristics (such as particle size and size distribution, morphology, material composition and density), operating conditions (i.e. moisture, temperature, static charge) and the current state of the powder (i.e. tapped, consolidated, aerated, free flowing etc.). The flow behaviour of a powder is brought about by a combination of the factors listed above; it is therefore difficult to characterise the flow properties in a universal way for all applications. Indeed, most flow characterisation techniques focus on specific aspects, such as measuring: flow rates through orifices of different size; the angle of repose; the energy to stir a powder bed; cohesion and internal angle of friction of the powder; bulk and tap densities; the formation of avalanches, etc. There is a vast amount of literature, patents, standards and specialised books and monographs dedicated to detailed descriptions of powder flow measurement methods, for example [5].

The above techniques are based on different principles and it has been recognised that the choice of flow characterisation technique should be made in relation to the process under investigation [6]. Yet, correlations have been found between techniques based on tapped density measurements, orifice flow, angle of repose and avalanching behaviour [6–8]. These techniques have been simplified and adapted to evaluate pharmaceutical powders, i.e. Refs. [9,10].

From a die fill point of view, however, Schneider et al., [11] have demonstrated that different flow measurement techniques

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can provide a different rank order of flowability. It is therefore important that flow characterisation is carried out using a device that captures the physical phenomena involved in die fill. Hopper flow has been investigated extensively and many of the flow characterisation methods described above are employed to assist hopper design. Flow into closed dies has, however, received less attention. In this paper we focus on powder flow through the delivery system of a rotary tablet press; more specifically on the flow mechanisms involved in die filling. We begin by describing the filling mechanisms of various tablet presses and examine a particular rotary press in detail. A model die–shoe filling system is used to capture the effects of gravity feed and the effect of air pressure on die fill. Experiments on typical pharmaceutical excipients and powder mixtures are presented. The flow is described qualitatively using high-speed video recording. A flowability metrics is introduced, which extends procedures developed to examine hopper flow [1] to account for the velocity of the feed shoe. The results are discussed in relation to the parameters of a rotary tablet press and the scaling of powder flow from the experimental system to the rotary press is analysed. This work builds on previous studies of die filling in the context of powder metallurgy and ceramic processing [12–14,11] and pharmaceutical powders [15].

2. The feed system of tablet production presses

The tablet compaction sequence typically includes die fill, tablet compression and ejection. Tablet presses can be divided into two broad categories: single station presses, which are driven by an eccentric cam, and multi-station presses, where the dies and punches are mounted on a rotating turret. We examine these machines from a die fill perspective.

In single station presses, the powder is fed into the die from a hopper and a feed shoe. The bottom punch is stationary during filling. The feed shoe is moved above the die opening, executes a number of shakes and is withdrawn. During this process, the powder is deposited into the die cavity. Depending on make or model, the powder movement in the feed shoe may be facilitated by a feeding mechanism (helical screws or rotating paddles, for example). The filling mechanism described above is very similar to typical presses employed in metal powder compaction, where a shoe executes a number of passes in order to fill the die. The filling process is essentially gravity assisted and the principal parameters are the shape and size of the feed shoe and feed shoe kinematics. For older presses, the mechanisms executing punch motion and die fill are usually mechanically interlinked and the shoe kinematics is dictated by the operating speed of the press. Modern presses allow for a more independent adjustment of fill parameters.

Single station presses are mostly used in research and development; while rotary presses are used for high volume production (i.e. hundreds of thousands of tablets per hour). Die fill on rotary presses is more complex. The operating cycle of a rotary tablet press is illustrated in Fig. 1. As the die Table 1 rotates as indicated in Fig. 1a, each tooling station passes successively through the die fill mechanism (4), compression rollers (6 and 7) and ejection cam mechanism (8). The characteristics of die fill are described below. The filling system consists of a mass flow hopper connected to a feed frame. In modern feed frames the powder is circulated using up to three rotating paddles to ensure uniform die fill and powder movement through the fill system.

In the following, we discuss the feed shoe of a Fette 1000 production press (Manufactured by FETTE GMBH, Schwarzenbek, Germany), to identify the feed mechanisms and the typical ranges for the operating parameters. The feed frame (Filo-matic type) consists of three wheels, for dispensing, feeding and metering, which are interlinked by interchangeable gear mechanisms; we consider the case where all wheels operate at the same speed. Each wheel is composed of 12 paddles.

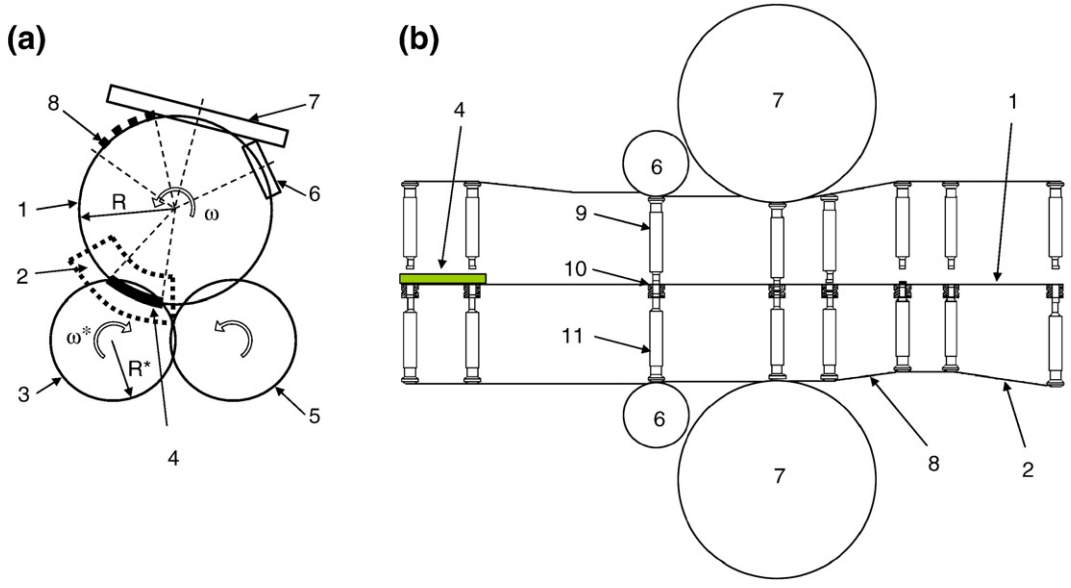


Fig. 1. Rotary press production cycle a) top view, b) unfolded view; 1 — die table, 2 — fill cam, 3 — feed wheel, 4 — die fill area, 5 — metering wheel, 6 — pre-compression roller, 7 — main compression roller, 8 — ejection cam, 9 — upper punch, 10 — die, 11 — lower punch.

Table 1
Characteristics and operating conditions of a rotary press feed system

	Symbol	Value	Example value
Press speed/rpm	ω	0–75	40
Feed frame speed/rpm	ω^*	0–100	20
Radius of die table (pitch radius) /mm	R	157	157
Effective radius[a] of feed wheel /mm	R^*	78	78
Angle where die receives powder /degree	α	35	35
Number of paddles of the feed wheel	N_S	16	16

[a] The effective radius is taken as the centre of a die. The nominal radius of the wheel is 100 mm.

We focus on the feed frame components above the die table (the feeding and metering wheel) and the fill cam and weight adjustment mechanism below the die table which are in the immediate vicinity of the die during the filling process as illustrated in Fig. 1a. The feed system is motor driven and the speed can be adjusted within the limits presented in Table 1. The table presents the geometrical characteristics of the system, which were measured, or taken from the press operating manual or drawings.

The paddle wheels rotate in the direction presented in Fig. 1a. While the die passes under the feed frame in the highlighted area, the die opening is exposed to powder and passes under a number of paddles (N_S) of the feed wheel. The number of passes can be calculated using elementary geometry as discussed in more detail elsewhere [15]:

$$N_P = \frac{N_S \alpha}{360}\left(\frac{R}{R^*} - \frac{\omega^*}{\omega}\right) \quad (1)$$

The symbols used in Eq. (1) are defined in Table 1 together with the range specific to the production press discussed.

The number of passes, N_P, is an important operating parameter because it can be related to the experimental system described in the following sections, where the number of passes necessary for complete filling of a model die can be established experimentally. If procedures for scaling the die fill process from the model die to a typical production size die can be established (i.e. the required value of N_P can be estimated) then the required speed of the feed frame can be obtained by solving Eq. (1) for ω. We discuss these aspects in more detail in Section 5.

Setting the press and feed frame speed to 40 rpm and 20 rpm respectively and using the parameters presented in Table 1, the resulting absolute velocity of the die is 657 mms^{-1}, while the absolute velocity of the paddles is 163 mms^{-1}in the same direction. This gives a relative velocity of the die with respect to the spokes of 494 mms^{-1}. For this situation we obtain N_P=2.35, i.e. there will be a little more than 2 passes of the spokes over the die cavity while the die is under the feeding zone.

It is possible to choose the press and feed frame speeds such that $\omega^* = \omega R / R^* \approx 2\omega$ which results in N_P being zero, i.e. there is no relative movement of the powder with respect to the die. In this case the contribution to die filling from the sweeping movement of the feeding wheel is reduced. Positive and negative values of N_P indicate that the relative velocity of the powder driven by the feeding wheel is either in the same, or the opposite, direction to the rotation of the die table.

This discussion, however, is relatively simplistic because it assumes gravity feed only, i.e. the material from the powder bed above the die table deposits in the die under the effect of gravity. In practice, additional complex phenomena take place, such as: force feeding, suction feeding, centrifugal forces and weight adjustment as described below.

The arms of the feeding and metering wheel can be profiled such that the powder is forced into the die cavity, this effect is referred to as force feed. The clearance between the paddles and die table (6 mm for the press examined here) suggests that there may be a zone where the powder is exposed to high shear conditions. The feed cam, which is shown as a dashed line in Fig. 1, is positioned such that the bottom punch has a vertical downwards movement, which starts prior to the empty die being exposed to powder in the feed-frame and continues while the die passes through the filling zone. This creates a negative pressure (vacuum) between the punch and the powder bed, facilitating the fill of the die through a suction effect.

An important difference between single station and rotary presses is the weight adjustment mechanism. In single station presses, the tablet weight is determined by the position of the bottom punch, which creates a fixed fill volume. For rotary presses, the weight adjustment mechanism is located after the bottom punch passes through the fill cam. Here the powder is moved upwards rapidly, essentially it is an overfill-eject excess mechanism. Then the top of the powder bed in the die passes under the metering wheel while the vertical velocity of the bottom punch is zero; in other words the volume of the die cavity is constant while passing under the metering wheel. It is unclear whether the metering wheel contributes to adding (forcing more powder into the die) or removing (through fluidization effects at high speed) material from the die. Production presses are designed not to transmit vibrations to the feed frame to avoid segregation, however, the inherent vibrations may influence the flow and packing of powder as it is transferred from the hopper to the feed frame and into the die.

The above analysis is based on a particular production press and feed system. It, however, illustrates the complex nature of the die fill phenomenon. The contribution of gravity feed, suction filling, centrifugal forces, weight adjustment, and the effect of the metering wheel, vibrations from the press and air pressure can however be studied separately.

In this paper we consider a simplified filling process in which powder is delivered into a die using a shoe. The position of the die is fixed and the shoe moves with a linear trajectory over the die opening. This system allows the effects of gravity feed and the relative motion of the delivery system and die to be assessed. Tests can be conducted in air or vacuum, allowing the effects of air flow and air pressure to be directly measured. By testing a range of powders that are typical of those employed in the pharmaceutical industry, the influence of different powder characteristics can also be assessed.

3. Experimental program

3.1. Materials

The powders selected for this study are either excipients or representative mixtures of excipients used in direct compression formulations. Seven powders were selected as presented in Table 2. Two grades of lactose were used: NF Lactose — 316 Fast Flo® (manufactured by Foremost Farms, USA) which is a monohydrate and NF Anhydrous Lactose — Direct Tabletting (manufactured by Quest international). For simplicity we refer to these as Lactose monohydrate and Lactose anhydrous, respectively.

Two grades of microcrystalline cellulose (MMC), Manufactured by FMC BioPolymer, Cork, Ireland were also tested: Avicel PH102, which is used as a direct compression excipient due to its good flow properties and Avicel PH101, which is of smaller particle size. These are referred to as coarse and fine microcrystalline cellulose (MCC). The fine grade was selected to allow comparison between the flow properties of the same powder material, but with different nominal particle sizes. Calcium Phosphate Dibasic Anhydrous, known also as A-TAB (manufactured by Rhodia) was chosen because of its ceramic like properties. Two powder mixtures having compositions typical of direct compression placebo formulations were also used. These mixtures consist of coarse MCC, one of the two grades of lactose; they also contain disintegrants, binders and lubricants as detailed in Table 2.

To ensure that the sample tested was representative of a batch of powder, it was remixed/reblended in a mixer (Turbula T2F, supplied by Willy A. Bachofen AG Maschinenfabrik Basel/Switzerland) prior to testing.

3.2. Die filling rig

The die–shoe filling rig used for this study consists of a pneumatically driven, microprocessor controlled rectangular shaped shoe, which translates over a fixed die. The shoe kinematics, i.e. acceleration and velocity, can be changed within a range representative of industrial processing. The entire system is located in a vacuum chamber. The influence of air on the filling process can be evaluated by comparing results from tests conducted in air and vacuum.

The present experiments were carried out in air and vacuum at shoe accelerations of 50 ms^{-2} and velocities in the range 0.01 to 1 ms^{-1}. Five shakes were applied to the shoe after it was filled and before it was translated over the die. The shakes consist of forward and backward motion at high velocity over a small displacement of ±5 mm. This is to create a reproducible initial state of the material within the shoe.

In all the experiments, the centre of the shoe, which is 30 mm wide and 60 mm long (in the direction of shoe movement) passes over the centre of the die to minimize the effect of any frictional interaction between the powder and shoe wall. High-speed video recording was used to characterize the flow behaviour qualitatively, as detailed in the following sections.

Experiments were conducted using a range of different dies, all in the form of rectangular parallelepipeds. The qualitative high-speed video studies were carried out using dies of opening 23 × 20 mm and 10 × 10 mm, both of height 38 mm. Quantitative studies were carried out with die openings of 10 × 10 mm, 15 × 15 mm or 20 × 20 mm, and heights of 70, 52, 38 or 20 mm. If not mentioned otherwise, the height of the powder in the shoe was kept constant at 18 (±2) mm for the smaller opening. This was increased to 27 mm for the intermediate opening and 35 mm for the largest die, to maintain similar types of flow processes. A major objective of increasing the die size is to determine how the results for a standard die can be extrapolated to other die profiles and filling conditions. Graphical procedures guided by dimensional analysis are presented in Section 4.1.

3.3. Types of flow

High-speed video has been employed to observe the details of the flow process as the powder is delivered into the die. Here we describe the general features of the flow process. More

Table 2
Powder properties

Powder	Description	Apparent density/ (g/ccm)	Tapped density/ (g/ccm)	Full density/ (g/cc)	Carr index	Hausner ratio	Nominal particle diameter/ micron
Placebo mixed powder 1	Lactose anhydrous, coarse MCC, croscarmellose sodium (disintegrant), magnesium stearate (lubricant).	0.52	0.72	N/A	28.57	1.40	N/A
Placebo mixed powder 2	Lactose monohydrate, coarse MCC, croscarmellose sodium (disintegrant), hydroxypropyl methylcellulose (binder), magnesium stearate (lubricant)	0.61	0.78	N/A	21.79	1.28	N/A
Calcium phosphate		0.78	0.82	2.89	4.88	1.05	180
Lactose monohydrate		0.58	0.67	1.540	13.43	1.16	60–120
Lactose anhydrous		0.68	0.81	1.589	16.05	1.19	100
Coarse MCC	PH102	0.34	0.48	1.512–1.668	29.50	1.42	100
Fine MCC	PH101	0.32	0.45	1.512–1.668	28.89	1.41	50

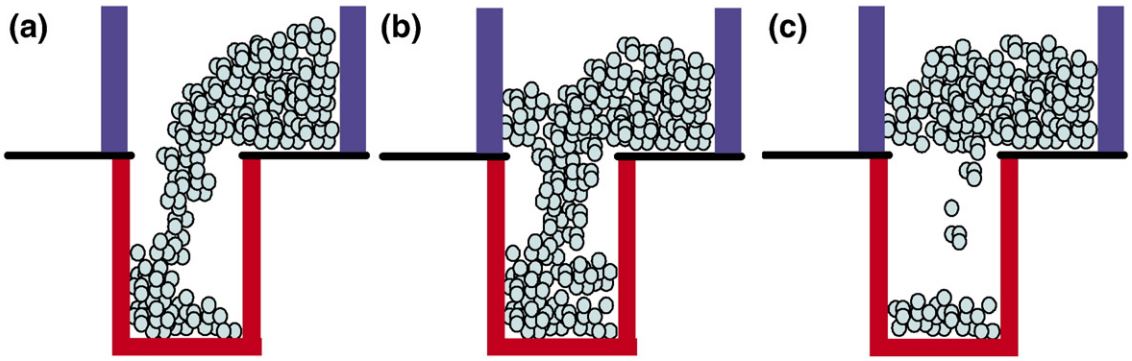


Fig. 2. Schematic representation of (a) nose flow, (b) bulk flow and (c) intermittent flow.

detailed information is presented elsewhere [13,15,11]. Three major types of flow have been identified: nose, bulk and intermittent flow, which are shown schematically in Fig. 2. When a fixed mass of powder is placed in a shoe, the initial acceleration of the shoe and friction between the powder mass and the base plate forces the powder towards the back of the shoe, forming a nose shaped profile. As the tip of this nose translates across the die opening, material can flow over the surface of the nose to the tip and avalanche into the die. We refer to this as nose flow, Fig. 2(a). At high speeds, or if the die opening is small, the tip of the nose rapidly moves across the opening and powder is delivered into the die by detaching from the bottom free surface of the powder mass. If this flow is reasonably continuous we refer to it as bulk flow, Fig. 2(b). For some materials, flow occurs as a result of a series of discrete instabilities, which releases either small clusters of particles or large chunks of agglomerated powder into the die. This is a random, often infrequent, process and we refer to it as intermittent flow, Fig. 2(c). In Fig. 2(c) the intermittent flow is illustrated as the random detachment of individual particles. Intermittent flow can, however, manifest itself in a number of different ways. Fig. 3 shows images taken from a video in which fine MCC powder was delivered into a die with an opening 20×23 mm. Cracks develop in the top of the powder mass as it translates across the die. As the powder becomes suspended across the die opening these cracks propagate towards the bottom, allowing large agglomerates to fall into the die. In contrast to freely flowing metallic and ceramic powders investigated elsewhere [11], pharmaceutical powders are more likely to flow intermittently. The following sections show that the development of a certain flow type depends on factors such as type of powder, shoe velocity, height of powder in the shoe, die opening size and whether the test is conducted in air or vacuum.

Fig. 4 shows a sequence of high-speed video images of the delivery of Placebo mixed powder 1 into a die of opening 10×10 mm. The experiments were conducted in vacuum at a steady state shoe velocity of 100 mms^{-1}. The time between images in the sequence in Fig. 4 is constant. Initially, nose flow is dominant and a fast rate of filling is observed, as illustrated in Fig. 4(a) to (c). As the shoe moves forward, the powder mass eventually completely covers the die; bulk flow is now the dominant filling mechanism and the filling rate decreases, see Fig. 4(d) to (f). In this experiment the initial height of powder in the shoe was sufficiently low to allow a considerable amount of nose flow to take place at the chosen shoe velocity. Since the centre of the shoe traverses over the centre of the die and the shoe is slightly wider than the die, we cannot observe the avalanche of particles from the top of the nose and into the die (Fig. 4(c)).

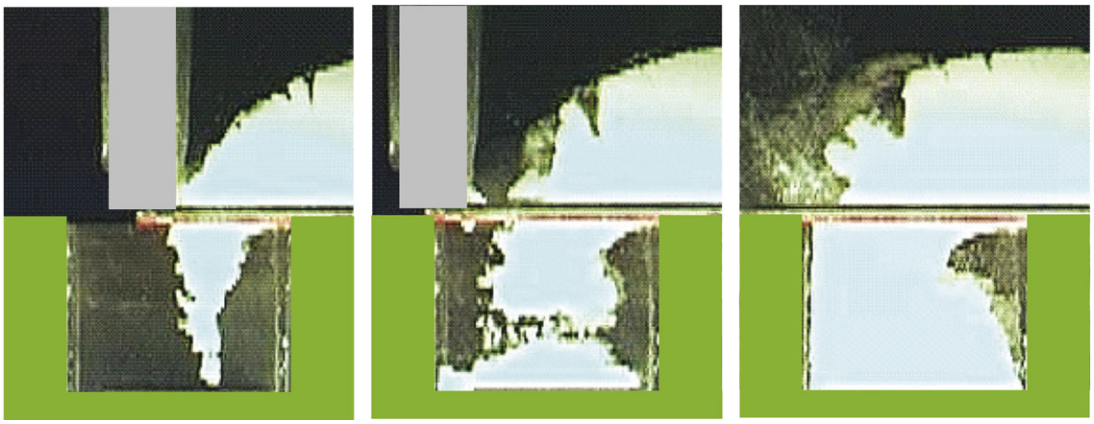


Fig. 3. Cracks at the top free surface of the powder mass propagate through the powder mass creating large blocks of material which fall into the die.

If the filling is conducted in air, air pressure builds up in the die as it is filled, which opposes the flow of powder, thus slowing down the rate at which it is delivered into the die. This can lead to a transition from bulk (Fig. 5(a)) to intermittent flow (Fig. 5(b)), as illustrated in Fig. 5 for Placebo mixed powder 1.

It is evident from these observations that the details of the flow process depend on the environment, the kinematics of the shoe motion, the die size and shape and the micro- and macroscopic properties of the powder. Any laboratory based technique for assessing the flowability of a powder must, therefore, be able to reproduce the major features of the flow mechanism that occur in the situation of practical interest.

3.4. Dimensional analysis

In a recent study of the flow of ceramic and metal powders Wu et al. [13] observed that the net flow rate, averaged over the entire delivery process, is a function of the shoe velocity. If the delivery system is stationary then the flow from a shoe is similar to that from a hopper. Hopper flow has received significant attention in the literature [16,1]. It proves instructive under these conditions to define a non-dimensional mass flow rate

$$\bar{M} = \frac{M}{\rho_a g^{1/2} D^{5/2}} \tag{2}$$

where ρ_a is the apparent density, g is the acceleration due to gravity and D is a characteristic dimension of the hopper opening. A number of studies have indicated that under conditions where core flow occurs $\bar{M}$ is constant for a given orifice shape, provided the height of powder in the hopper is at least twice D. This form of relationship was confirmed by Nedderman [1] using dimensional analysis, if it is assumed that the discharge is only a function of ρ_a, g and D and is weakly dependent on friction. In this section we extend the dimensional analysis of Nedderman [1] to take into account the motion of the shoe. At the same time we explore the effect of other factors on the flow of the powder.

Consider the situation where a die is filled in vacuum at a steady state shoe velocity, v_s, and a single mechanism dominates the filling process. Compared to Nedderman's analysis [1] the only additional quantity that needs to be included is the shoe velocity (although there are two potential coefficients of friction, that for sliding between the particles, μ_p, and that for sliding between the particles and base plate, μ_{bp}). Now, the mass flow rate is a function of ρ_a, g, D, v_s, μ_p, μ_{bp} and the shape of the die opening (and its orientation with respect to the direction of shoe motion), i.e.

$$M = f(\rho_a, g, D, v_S, \mu_p, \mu_{bp}, \text{shape}) \tag{3}$$

or $$F(M, \rho_a, g, D, v_S, \mu_p, \mu_{bp}, \text{shape}) = 0$$

Taking ρ_a, g and D as the independent variables and employing the Buckingham Π theorem [17] we obtain two independent dimensionless groups, $\bar{M}$ given by Eq. (2) and

$$\bar{v} = \frac{v_S}{g^{1/2} D^{1/2}} \tag{4}$$

We can now write Eq. (3) in the form

$$\bar{M} = f(\bar{v}, \mu_p, \mu_{bp}, \text{shape}) \tag{5}$$

For a given powder, base plate and die profile $\bar{M}=f(\bar{v})$. Over a range of velocities it is possible to fit a power law relationship to the data, i.e.

$$\bar{M} = C\,\bar{v}^{\,-n} \tag{6}$$

where C and n depend on the powder properties, through μ_p and μ_{bp}, and the shape of the die opening.

We use Eq. (6) to guide the interpretation and correlation of experimental results. In the derivation of Eq. (6) we considered

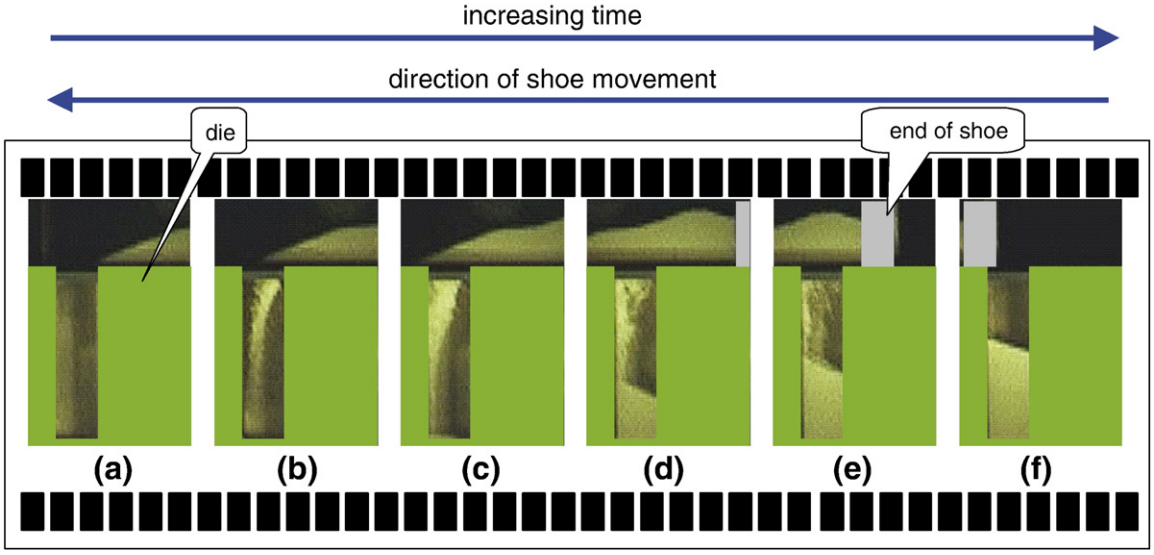


Fig. 4. High-speed video snapshots taken during the delivery of Placebo mixed powder 1 into a die of opening 10×10 mm at a shoe speed of 100 mm/s. Initially, fast nose flow takes places, see (a) to (c). When the powder completely covers the die the flow rate reduces, since bulk flow becomes the dominant mechanism, see (d) to (f).

the situation where the powder is delivered into the die in vacuum. A number of additional factors come into play when the powder is delivered in air [13]. As the powder mass dilates as it starts to fall into the die a negative pressure gradient is created which acts to oppose the flow of powder [18]. Also, as the amount of powder in the die increases, air pressure can build up in the die, which enhances the magnitude of the pressure gradient and further slows down the powder flow rate. During a particular delivery process the flow rate therefore decreases. The magnitude of the pressure gradient will depend on the permeability of the powder bed, which is a function of the fluid viscosity, η, and the size, d, and shape of the particles. Using these fundamental quantities provides two additional dimensionless groups:

$$\bar{\eta} = \frac{\eta}{\rho_a g^{1/2} D^{3/2}} \qquad \text{and} \qquad \bar{d} = \frac{d}{D}$$

When examining the effect of air on the powder flow process we examine the net flow rate over the filling period. For a given die geometry $\bar{\eta}$ and $\bar{d}$ are constant and it is possible to correlate the experimental results using Eq. (6) but C and n are now functions of $\bar{\eta}$ and $\bar{d}$. There is a significant difference between the values of $\bar{\eta}$ for tests conducted in air and vacuum, and we would expect to observe different values of C and n. The influence of air and particle size on the filling process can therefore be assessed by comparing the values of these quantities from tests in the two different environments.

It is evident that the results in vacuum could also be sensitive to the value of $\bar{d}$. In studies of hopper flow [16,1] the results are generally expressed in terms of an effective die diameter $D_{\text{eff}}=D(1-1.4\bar{d})$, which should replace D in Eqs. (2) to (4). For the range of powders considered here, this correction is not significant and throughout the remainder of this paper we employ the form of equations presented above.

In the above discussion we have assumed that the powder is cohesionless and flows freely. For some powders, other effects can come into play and these can be evaluated using the framework described above. For example, if there are attractive cohesive forces between the particles other dimensional groups can become important. If we can characterise the cohesive properties in terms of the cohesive strength of the powder, σ_c, then the flow behaviour is influenced by the dimensionless strength

$$\bar{\sigma}_c = \frac{\sigma_c}{\rho_a g D} \tag{7}$$

This parameter can become significant when the die opening is small. Thus cohesive strength can play an important role in determining the flow behaviour into dies with small openings and can influence whether the flow is smooth or intermittent.

3.5. Critical velocity

The rig described in Section 3.2 can be used to determine the mass of powder in the die as a function of shoe velocity. It is then possible to determine the fraction of the die that is filled as a function of shoe velocity. Wu et al. [13] demonstrate that a useful measure of the flowability of a powder is the critical velocity v_c, which is the shoe velocity above which incomplete filling occurs for a single pass over the die. It depends on the type of powder, the shoe kinematics, including the number and intensity of any shakes, shoe and die design, environment (air/vacuum) and the amount of powder in the shoe. Detailed investigations of metallic, ceramic and magnetic powders, which examine the influence of each of these variables, are presented elsewhere [13,11]. Here we provide a stronger fundamental basis for use of the critical velocity based on the dimensional analysis presented above. The analysis also suggests ways in which the results from a series of experiments can be extrapolated to other die and shoe geometries.

Consider the situation where a shoe of length L moves over a die of uniform cross-section and height H, dropping powder into it. Assuming a constant mass flow rate into the die over the delivery period allows the time required to completely fill the die to be determined and the critical velocity, v_c, above which the die is not completely filled. Using Eq. (6), we find

$$v_c = \sqrt{gD}\left(C\frac{L}{H}\right)^{\frac{1}{1+n}} \tag{8}$$

and for velocities higher than this the fraction of the die that is filled, δ, is given by

$$\delta = \left(\frac{v_c}{v_S}\right)^{1+n} \tag{9}$$

From a practical perspective, it can be more convenient to characterise the flowability of a powder in terms of the critical velocity and the exponent n for a standard die and to use Eqs. (8) and (9) to predict the response for other die geometries. Extrapolation of the results in this way can only be undertaken if the flow mechanism remains the same. We explore this more fully in the following sections.

When comparing results generated using different die profiles, it is more convenient to express the results in terms of the mass m of powder delivered to the die. Now

$$\bar{m} = \frac{m}{\rho_a D^2 L} = C\left(\frac{v}{\sqrt{gD}}\right)^{-(1+n)} = C\,\bar{v}^{\,-(1+n)} \tag{10}$$

For experiments in vacuum, this expression allows the flow behaviour to be uniquely characterised, since C and n are independent of die geometry.

4. Interpretation of experimental results

In this section we use the analyses of Sections 3.4 and 3.5 to provide a framework for the interpretation of the experimental results generated for the powders listed in Table 2.

4.1. Mass flow rate as a function of shoe velocity

For a given flow mechanism, Eq. (10) provides a relationship between the normalised net mass flow rate, $\bar{m}$, which has been

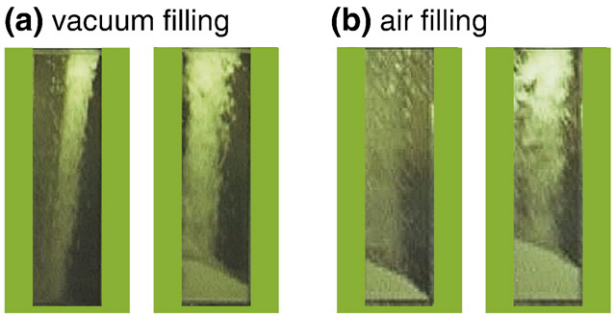


Fig. 5. Flow of Placebo mixed powder 1 into a simple die in (a) vacuum and (b) air.

delivered to the die as a function of the normalized shoe velocity, $\bar{v}$, given by Eq. (4). This equation can only be employed if the die is not completely filled during a single sweep of the shoe. Results from a series of experiments on the Placebo mixed powder 1 using dies of different geometry are plotted in Fig. 6. This figure shows results generated in air and vacuum. Open symbols correspond to tests conducted in vacuum. In all these experiments the powder was delivered to the die by a combination of nose and bulk flow. All other symbols are for experiments in air: solid symbols indicate that delivery was by a combination of nose and bulk flow, while half filled symbols indicate that intermittent flow dominated the delivery process. Triangles correspond to experiments for the reference die ($10\times10\times38$ mm).

4.1.1. Filling in vacuum

The experiments in vacuum, see open symbols in Fig. 5, were carried out using dies with 10×10 mm^2 (small) and 20×20 mm^2 (large) openings, each with two different heights. The two heights for the die with a small opening were 38 mm (open triangles) and 70 mm (open circles). The larger opening size was investigated with heights of 52 mm (open square) and 70 mm (open diamond). It is evident from Fig. 5 that the results in vacuum show a consistent trend, which can be correlated using Eq. (9). Fitting this expression to the data of both die openings at different heights we obtain $C=0.07$ and $n=0.25$.

4.1.2. Filling in air

Experiments in air, see filled and half filled symbols in Fig. 6, were carried out using the small, medium and large die openings. The data obtained with the large die opening for heights of 38, 52 and 70 mm are shown together as solid circles (Fig. 6); indicating that for each of these dies filling takes place by a combination of nose and bulk flow. Intermittent flow is, however, observed when filling the small and medium sized dies; the data are presented as half filled triangles and half filled squares, respectively. The results for the tests conducted in air are spread over a wider range than those in vacuum. At a given normalized velocity the mass delivered to the die is always less than that in vacuum and intermittent flow always provides a slower filling rate than nose/bulk flow. Intermittent flow dominates for the smaller die at slow shoe velocities. There is significantly more scatter in the experimental results when intermittent flow occurs, since this is a random process in which large agglomerates detach and fall into the die, separated by periods of negligible flow; under identical delivery conditions either zero or several agglomerates can fall into the die.

A trend line has been fit to the air data for all die openings and depths in Fig. 6. This line lies about one order of magnitude below that for the tests in vacuum, indicating a strong effect of air on the filling process. The scatter in results does not allow us to fully evaluate the influence of die opening (or the dimensionless quantities d and η introduced in Section 3.4) on the filling process, but we can identify some interesting trends in behaviour. In the interpretation of the results it is important to recognise that the balance between nose and bulk flow changes as the normalised velocity increases. At slow shoe velocities, i.e. low values of $\bar{v}$, nose flow dominates while at higher velocities bulk flow is the dominant flow mechanism.

At slow shoe velocities we observe a trend in which the normalised mass delivered to the die decreases with decreasing die opening, i.e. nose flow provides a faster filling rate than the intermittent processes observed when using a smaller die. Also, cohesion between the particles becomes more important in determining the flow into small dies (see Eq. (7)). This could explain the more intermittent flow process in these situations,

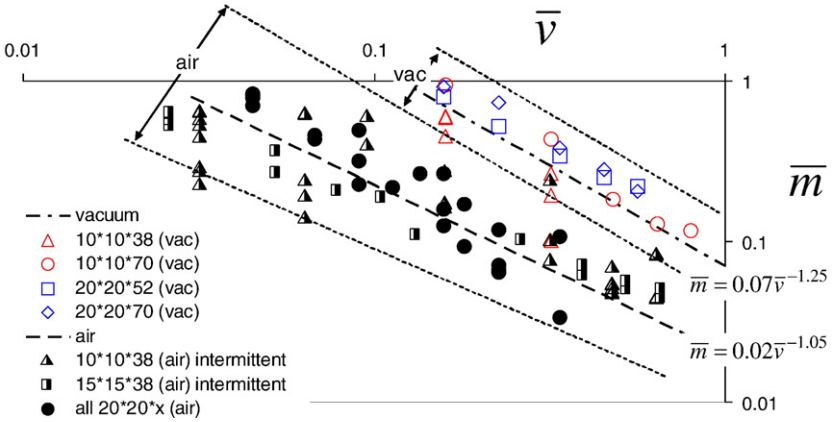


Fig. 6. Normalised logarithmic plot of the mass of powder delivered to a die as a function of velocity for Placebo mixed powder 1 with different die geometries. Tests in vacuum are represented by open symbols. The other symbols represent the tests conducted in vacuum: solid symbols indicate bulk and nose flow; and half filled symbols represent intermittent flow.

although we would also expect an influence of cohesive strength in vacuum, unless the environment (air or vacuum) has an influence on the cohesive properties. It is evident from the images of Fig. 3 that powders tested in this study have some cohesion, but we cannot determine its full influence through the range of tests conducted here.

At higher shoe velocities we observe the opposite trend, i.e., $\overline{m}$ increases as the die opening decreases. At these velocities particles detach from the bottom of the surface of the powder mass which completely covers the die. Independent of whether this is a smooth (bulk flow) or intermittent process, the build-up of pressure in the die is determined by the permeability of the powder bed. Micromechanical models of the flow of air through an array of particles, suggest that the permeability is proportional to d^2/η [21]. Since the difference between the behaviour in air and in vacuum is largely determined by the permeability of the flowing powder, this suggests that C and n are functions of

$$\overline{d}^{\,2}\overline{\eta} = \frac{\rho_a g^{\frac{1}{2}} d^2}{\eta D^{\frac{1}{2}}} \tag{10}$$

For a given powder ρ_a and d are constant. Also taking g and η as constant, we find that the normalized permeability is proportional to $D^{-1/2}$. In this regime, we would expect a higher pressure to build-up in the die for a lower normalised permeability, i.e. a larger die opening, and therefore a slower rate of delivery of powder into the die. This is consistent with the experimental observation (Fig. 6) at high velocities, but because of the scatter in experimental data and the small range of values of D used in this study it is not possible to accurately determine how the flow rate depends on d^2/η or D.

The values of C and n for Placebo mixed powder 1 were determined in vacuum using four different die geometries. Using expression (8), the critical velocity can now be calculated for other die geometries. As an example, C and n in vacuum substituted into expression (8) for the reference die geometry ($10\times10\times38$ mm) and a shoe length of 60 mm gives $v_c = 54$ mms^{-1} which compares well to the critical velocity of 45 mms^{-1} measured for this die.

Full verification of the dimensional analysis presented here requires evaluating a range of powders, which lies outside the purpose of this study. The detailed experimental results for the placebo mix presented in Fig. 6 nevertheless demonstrate that Eq. (10), and therefore Eqs. (8) and (9), can be used to describe the flow behaviour during die fill for a typical pharmaceutical powder system.

4.2. Fill ratio velocity relationship

When comparing the behaviour of different powders it is more useful to characterize the response in terms of the critical velocity defined in Eq. (8). In practice, the fill ratio δ can be readily determined experimentally. Fig. 7 provides plots of δ against v_S for two powders using a die with an opening (10×10 mm) and a height of 38 mm. At slow shoe velocities the die completely fills and $\delta = 1$. At higher velocities filling is incomplete and δ decreases sharply with increasing velocity. In Fig. 7 the critical velocity is given as the intersection of the horizontal straight line, which

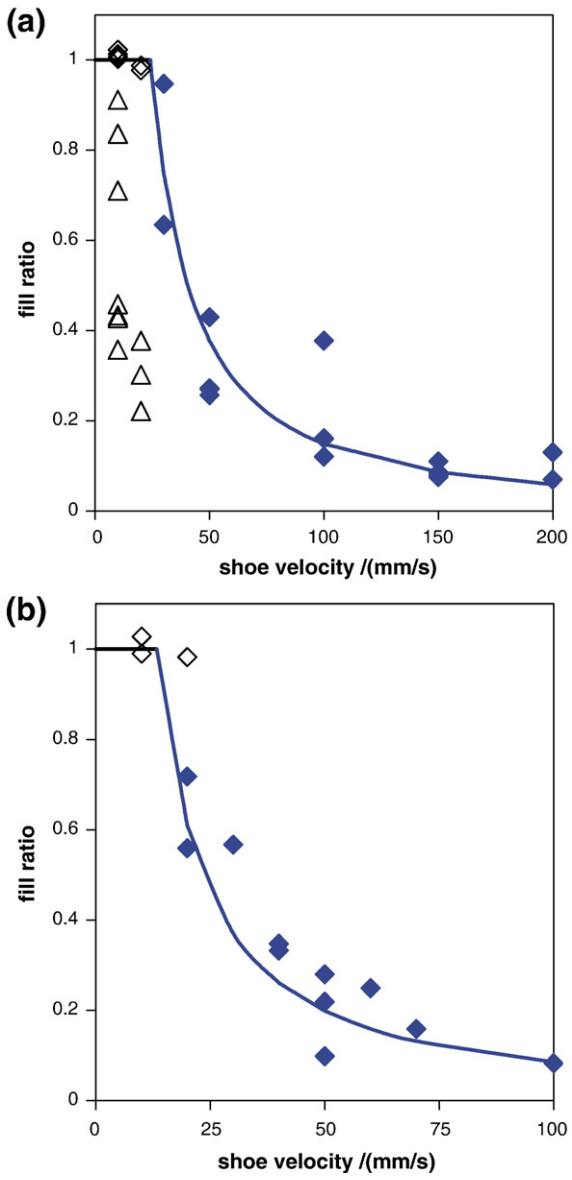


Fig. 7. Determination of critical velocity in air with n as a free parameter using Eq. (8) to fit the data for (a) Placebo mixed powder 1 and (b) coarse MCC. The data points labelled with triangles indicate intermittent flow regimes and were not included in the determination of the critical velocity.

passes through unity on the ordinate, with the curved line at higher velocities, which is given by the fit to the data using Eq. (9). Sinka et al. [15] show that the scatter in the fill ratio–velocity relationship is large if intermittent flow takes place. The scatter in results, however, is less if the delivery process is smooth, i.e. if bulk and/or nose flow dominate. Fig. 7(a) shows the intermittently flowing Placebo mixed powder 1 in air. Data points corresponding to intermittent flow are shown as triangles. These data points were not used when fitting Eq. (9). The critical velocity determined when not taking into account all points, i.e. omitting intermittently flowing powder, is therefore a maximum possible critical velocity. Sinka et al. [15] show that the flow of the coarse MCC powder is much smoother and continuous in air, see Fig. 7 (b), resulting in a greater reproducibility of the experimental results.

Similar plots to those shown in Fig. 7 were constructed for each powder and the critical velocities were determined using the procedure described above. It was not possible to determine critical velocities using the current die for the poor flowing powders, i.e. fine MCC and the Lactose anhydrous powder. Even at the slowest possible shoe velocity of 10 mms^{-1}, or just by opening a shutter below the shoe while it is located over the die, it was not possible to completely fill the die.

Values of the fitting parameter n are given in Fig. 8 for all the powders tested. These values are in the same range as those obtained from previous studies on metallic, ceramic and magnetic materials [13,11]. In these studies on other powders it was observed that the critical velocity is not sensitive to the value of n used in the fitting procedure and use of $n=0.2$ gives a good fit for a wide range of powders. Fig. 9 shows the critical velocities determined by either treating n as a fitting parameter or specifying its value as 0.2. It is evident that similar values of the critical velocity are obtained for each powder, despite the spread in values of n which provide the best fit (Fig. 8). Use of a fixed value of n simplifies the fitting procedure and amount of experimental data required in determining the critical velocity [11].

The flowability of pharmaceutical powders can strongly depend on air temperature and relative humidity. For the set of experiments carried out here the temperature and relative humidity were recorded, but their influence on the critical velocity was found to be smaller than the scatter in experimental results. For the Placebo mixed powder 1, the fill ratio–velocity relationship was investigated in more detail. Experiments were carried out at relative humidities of 30 to 65% and temperatures between 10 and 25 °C. In order to reduce the moisture content, the powder was either evacuated for 20 min in a vacuum better than 10 mbar or heated at 100 °C for 2 h immediately prior to being tested. Intermittent flow was observed in all cases and neither the critical velocity nor the scatter in fill ratio for a given shoe velocity was changed if the sample was dried by one of the methods mentioned above or not.

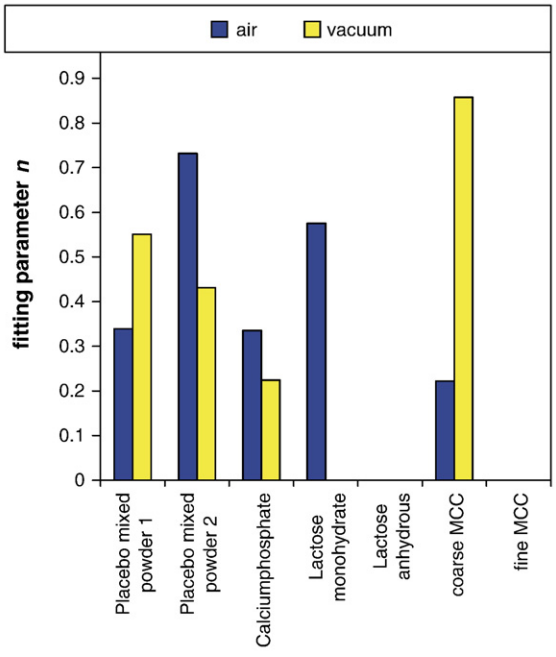


Fig. 8. Range of fitting parameter n from expression (8) in air and vacuum. Data corresponding to intermittent flow are not considered.

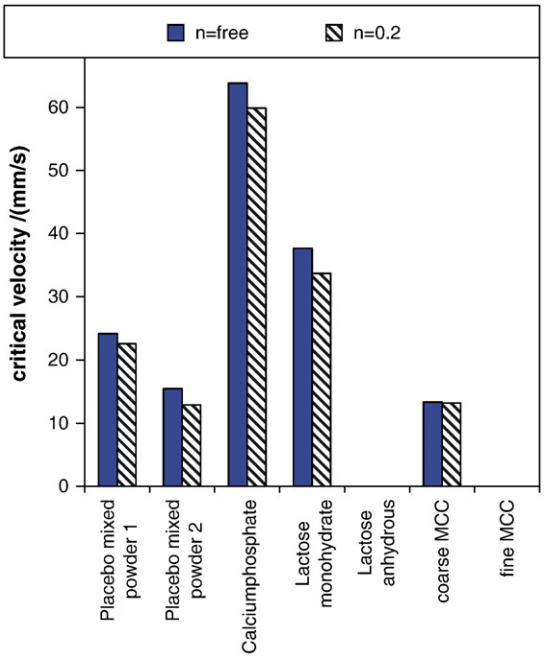


Fig. 9. Influence of parameter n on the critical velocity for regimes excluding intermittent flow.

4.3. Influence of environment on fill density and critical velocity

Schneider et al. [11] showed that for a range of different powders the environment, i.e. if filling takes place in air or vacuum, systematically influences the fill density. Fig. 10 shows that the influence of environment on fill density and apparent density is entirely material dependent. For metallic, ceramic and magnetic powders it is observed that higher fill densities are obtained in vacuum and the apparent density (as determined using a standard flowmeter) is lower than the fill densities in both environments [11]. This trend is not evident for the pharmaceutical powders tested here.

The environment has an influence not only on the fill density, but also on the critical velocity, as can be seen in Fig. 11, which is generally significantly higher when filling the die in vacuum. All critical velocities in this figure were determined with n as a free fitting parameter. From the high-speed video recordings smoother, i.e. less intermittent, flows are generally observed in vacuum as discussed in Section 3.3, which is due to the absence of the build up of air pressure in the die opposing the flow process. Fig. 11 also shows a comparison of the Carr index and Hausner ratio with the two critical velocities determined in air and vacuum. The Carr index and Hausner ratio are commonly used measures for describing flowability, particularly in the pharmaceutical industry, but no correlation with the critical velocity can be seen (Fig. 11). It is evident that these different techniques provide different measures of the flow characteristics of the powders tested in

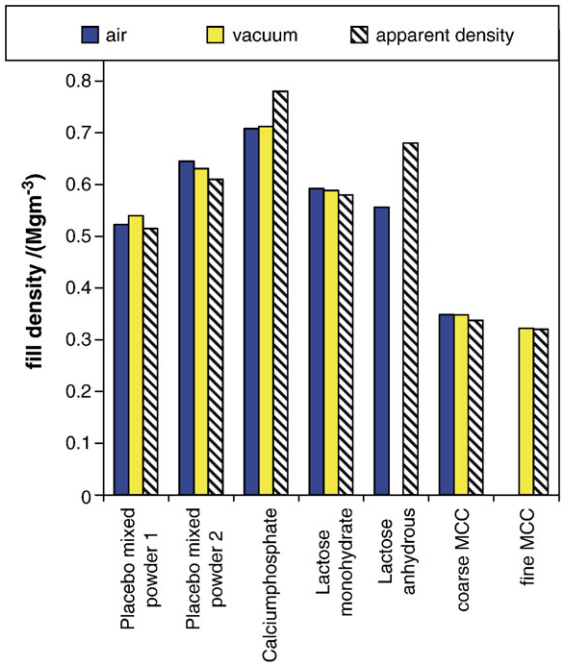


Fig. 10. Influence of environment on fill density and comparison to the apparent density.

this study. The shoe delivery system more closely resembles the type of processes the powder experiences during the delivery stage of a rotary press and it might therefore be expected to provide a more suitable measure of flowability.

5. Relationship between present results and flow in a rotary press

In this section, we use the results presented above to make a preliminary evaluation of the filling of a die using the rotary press described in Section 2. We pose the following question: "Can experiments used to determine the flowability of a powder be used to determine the maximum rotational speed at which a die can be completely filled in a rotary press?".

The rotary press examined in Section 2 employs suction filling. During suction filling the motion of the lower punch creates a vacuum under the powder mass and the positive pressure exerted by the air above the powder helps to deliver the powder into the die. During this process, air does not have to permeate through the powder mass and, as in the type of vacuum filling considered in this paper, we would therefore expect C and n of Eq. (6) to be independent of the size of the die opening. The positive air pressure above the powder experienced in the press provides an additional driving force for the delivery process and C for this situation could be greater than that determined using the shoe delivery system. C and n can also depend on the balance between nose and bulk flow and therefore on the height of powder in the delivery device. Under the conditions considered here bulk flow dominates the delivery process and the height of powder does not significantly influence the rate of delivery, provided it is at least twice the die opening [11]. This should also be the case on the press. As a first approximation we assume that C and n determined from the tests in vacuum are appropriate for the conditions experienced in the press. We also interpret velocity as the relative velocity between powder in the feed frame and the die and ignore the effects of centrifugal forces on the delivery process.

With C and n determined for vacuum in Section 4.1 the max press speed can now be calculated by substituting expressions (2) and (4) into (6) . The required minimum mass flow rate to fill the die completely is given by

$$M = \frac{\rho_a D^2 h}{\alpha/\omega} \tag{11}$$

where $\rho_a D^2 h$ is the mass to fill the die completely and α/ω the corresponding time of delivery obtained from the press geometry. The relative velocity v_r between die table and feed wheel is represented by the shoe velocity v_S in the die–shoe model rig; it can be expressed as

$$v_r = v_S = R\omega\left(1-\frac{R^*\omega^*}{R\omega}\right) \tag{12}$$

With the assumption that the ratio of radial velocities of the feed wheel and the die table, $R^*\omega^*/R\omega = k$, stays constant, the press speed is given by

$$\omega = \sqrt{\frac{gD}{R}}\left(\frac{CR\alpha}{h}\right)^{\frac{1}{1+n}}(1-k)^{\frac{-n}{1+n}} \tag{13}$$

Using the typical press values given in Table 1 and, as an example, assuming a die opening of 10 × 10 mm and a die height of 7.5 mm the maximum rotational speed of the die table ω

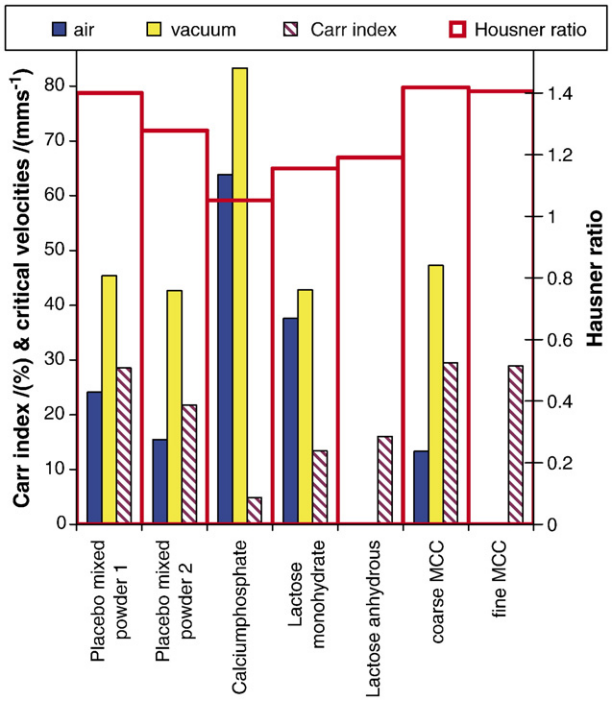


Fig. 11. Comparison of critical velocity in air and vacuum (for non-intermittent flow) with the Carr index and Hausner ratio. Die and shoe geometry and initial powder height in the shoe were kept constant.

becomes 21 rpm. This value is approximately half the press speed typically used in industrial processing, see Table 1. This can be due to a number of factors: use of values of C and n determined from vacuum filling could significantly underestimate the enhanced effect of suction filling. In more recent work Jackson et al. [19] have modified the shoe–die system in order to investigate the effect of suction. Results for coarse microcrystalline cellulose indicate that the critical velocity is increased by a factor of approximately 2.5 if suction filling is applied. Using of the results of this study now results in a predicted maximum rotational speed of the press that is consistent with that employed in practice. However, more detailed studies are necessary to scale the kinematics of suction feed and to include the force feed effect in order to extrapolate the experimental results from the experimental shoe–die system to tablet presses.

Of the systems that are available for measuring the flowability of powders, the shoe delivery system employed here comes closest to mimicking the type of conditions that the powder experiences in a rotary press. It was originally designed to model the gravity filling process commonly employed in the powder metallurgy and ceramics industries. The system does not, however, replicate important features of the rotary press delivery process that can play a significant role in determining the flowability of a powder. For pharmaceutical powders, suction effects are important and if the results of a flow test are to be used in a predictive manner, it is important that the system employed incorporates this feature of the delivery process. Use of suction does not, however, influence the flowability of powders with large (greater than 150 μm) heavy particles such as steel and hard metal powders [20] and for these powders the shoe delivery system can provide a good measure of the flowability under a range of filling conditions. These observations reinforce those of Wu and Cocks [14] that different techniques for measuring the flowability of a powder can give different rank orders of the flowability and the most appropriate technique for a given application is that which models those features of the process that have the most significant influence on the flowability in the situation of interest. There is no unique test and the results from a particular test should be used with care.

6. Conclusions

In this paper we have evaluated the flowability of seven pharmaceutical powders in the context of the delivery process in single station and rotary presses. High-speed videos of the delivery process in a model die–shoe filling rig has allowed the details of the flow process to be observed and the dominant mechanism for a given powder, environment, geometry and shoe kinematics to be identified.

A number of dimensionless groups have been identified, which provide a framework for presenting the results of a wide range of different experiments and identifying correlations within the data set. Two dimensionless parameters can be identified to correlate the flow behaviour of cohesionless powders in vacuum for a given flow mechanism: a dimensionless mass flow rate $\bar{M}$, defined in Eq. (2); and a dimensionless velocity $\bar{v}$, Eq. (4). Experimentally, it is observed that a simple power–law relationship is satisfied between these two parameters for a reasonably wide range of flow conditions. This result has provided further support for use of the concept of a critical velocity, originally proposed by Wu et al. [13], to characterise the flow behaviour of powders.

If the powder is delivered in air, and or there is some cohesion between the particles, a number of other dimensionless groups contribute to determining the flow behaviour. A number of groups have been identified here, but others could be important, depending on the details of the flow process and the exact state and composition of the powder. In situations involving suction filling, force feeding and fluidisation, other dimensionless properties are important and need to be considered when extrapolating the results presented here to situations where these effects become significant. The dimensional analysis presented here provides a framework for evaluating these other factors. Further work is, however, required to more fully evaluate the flow characteristics of particulate materials and to develop methods to extrapolate results beyond the precise conditions employed in laboratory experiments. The analyses presented here, none-the-less, provide a useful framework for the conduction of further studies and for developing sound extrapolation procedures.

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